PECARN Paediatric Head Injury Rule — CT Decision

children <18 y, GCS 14–15, within 24 h of head trauma

PECARN

RISK GROUPS AND CITBI RISK

Group	Age <2 years	Age ≥2 years	Action
High	GCS 14 or other altered mental status, OR palpable skull fracture	GCS 14 or other altered mental status, OR signs of basal skull fracture	CT recommended (ciTBI ≈4.4% / 4.3%)
Intermediate	Non-frontal scalp haematoma, LOC ≥5 s, severe mechanism, or not acting normally per parent	Any LOC, vomiting, severe mechanism, or severe headache	Observation vs CT (ciTBI ≈0.9%) — CT more likely if several findings, worsening, age <3 months, parental preference
Low	None of the above	None of the above	CT not recommended (ciTBI <0.02% / <0.05%)

SEVERE MECHANISM

- MVC with ejection, death of another passenger or rollover
- Pedestrian or unhelmeted cyclist struck by a motorised vehicle
- Fall >0.9 m (3 ft) if <2 y; >1.5 m (5 ft) if ≥2 y
- Head struck by a high-impact object

SA GUIDELINES

- TBI: CT within 0–6 hours; refer all children with TBI ✚
- Major-injury indicators include decreased level of consciousness; children <2 years with these require admission ✚
- Moderate head injury = GCS 9–12 ‡ ✚
- Severe TBI = GCS <9 → refer to a specialised centre ✚

PEARLS

- Applies only to GCS 14–15; GCS ≤13 → CT/referral regardless
- ciTBI = death, neurosurgery, intubation >24 h or admission ≥2 nights for TBI
- Isolated vomiting as the only finding (≥2 y): observation is reasonable
- Suspected non-accidental injury: rule does not apply — act on safeguarding

PAED STG & EML Paediatric Hospital Level, 2025-2029 cycle (updated Mar 2026) — pp 1.30, 23.22 · **PHC** STG & EML Primary Healthcare, 8th ed. 2024 (updated Sept 2026) — pp 21.63 · **PECARN** Kuppermann N et al. PECARN head injury rules. Lancet 2009;374:1160-70

✚ = SA STG / national guideline (pages listed above). Unmarked content: international source named above. ‡ = STG misprint or sources differ (see app). ANDH · pecarn · updated 2026-10-14